Scale-Dependent Dimensionality Reveals a Common Structural Motif Across Representational Systems

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Abstract

Understanding how high-dimensional representations organize across scales is a central problem in neuroscience and complex systems. Here, we analyze effective dimensionality as a function of scale across model-derived cortical representations and multiple synthetic systems. We find that effective dimensionality follows a consistent growth–saturation profile: locally low-dimensional structure expands with increasing scale before reaching a bounded plateau.

After normalization, dimensionality curves exhibit strong cross-domain alignment (mean similarity ≈ 0.90), indicating a shared structural motif. However, systems differ quantitatively in growth rate and saturation level. TRIBE-derived cortical representations show pronounced compression, with low effective dimensionality ($D_{\text{eff}} \approx 3$–5) and early saturation relative to other systems.

Extensive negative controls–including dynamical torsion, curvature, and residual structure–fail to reproduce robust alignment, indicating that the observed pattern is not attributable to specific dynamical artifacts. These results identify scale-dependent dimensional growth with bounded saturation as a general property of structured representational systems, while highlighting system-specific constraints on representational capacity.

1 Introduction

Characterizing the geometry of high-dimensional representations is a fundamental challenge across neuroscience, machine learning, and complex systems. In cortical organization, large-scale gradients provide a low-dimensional embedding of functional structure [Margulies et al., 2016], yet the multi-scale organization of representational geometry remains poorly understood.

Most analyses focus on fixed-scale properties, overlooking how local structure expands into global organization. A scale-dependent perspective is required to capture this transition.

Here, we analyze effective dimensionality as a function of neighborhood scale across multiple systems. This allows us to test whether representational systems share common structural patterns and to quantify how they differ.

2 Methods

2.1 Datasets

We analyzed:

- TRIBE v2 model-derived cortical representations
- Synthetic datasets: Gaussian, uniform, hierarchical, correlated, and sparse systems

All datasets were matched in sample size and dimensionality where appropriate.

2.2 Effective Dimensionality

Effective dimensionality was computed using the participation ratio:

$$D_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \tag{1}$$

where λ_i are eigenvalues of the covariance matrix.

2.3 Scale Dependence

Neighborhoods of size k were constructed, and $D_{\text{eff}}(k)$ was computed across a wide range of scales ($k = 5$ to 1000).

2.4 Normalization

To compare across systems:

$$\tilde{D}(k) = \frac{D_{\text{eff}}(k)}{D_{\max}} \tag{2}$$

$$\tilde{k} = \frac{k}{k_{\max}} \tag{3}$$

2.5 Curve Alignment

Similarity between normalized curves was quantified via correlation-based alignment.

3 Results

3.1 Scale-Dependent Dimensional Growth

All systems exhibit increasing effective dimensionality with scale (Fig 1). Local neighborhoods are low-dimensional, while larger neighborhoods reveal higher-dimensional structure.

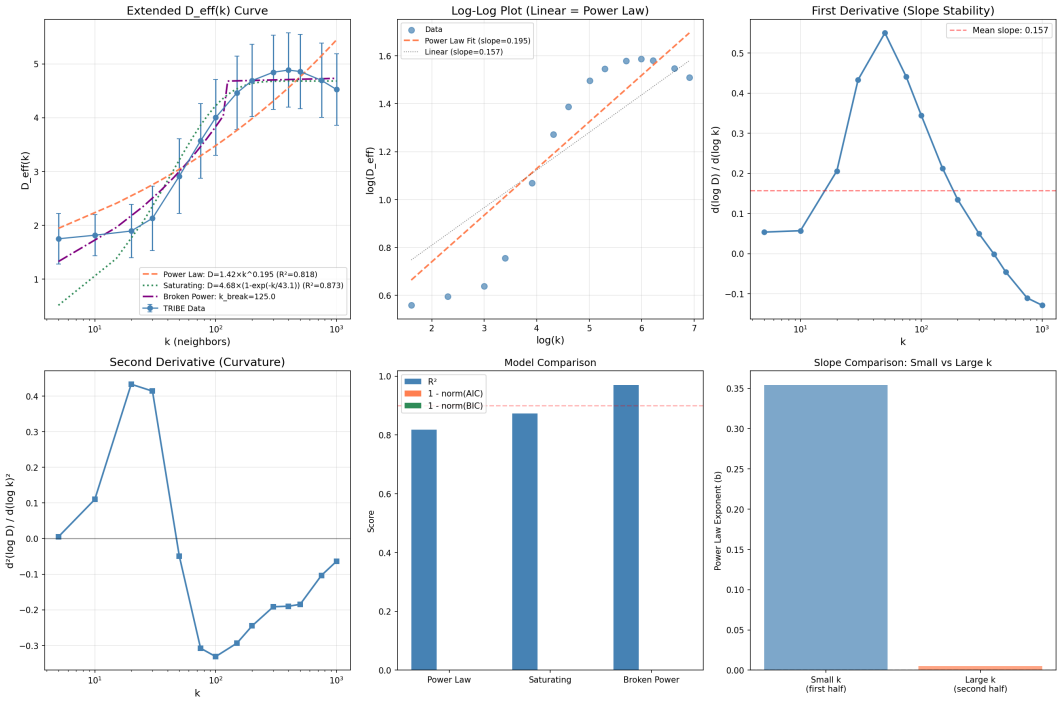

Figure 1: Effective dimensionality $D_{\text{eff}}(k)$ across scales for TRIBE and synthetic datasets. TRIBE exhibits rapid early growth followed by early saturation, while synthetic systems grow more gradually and saturate later.

3.2 Growth–Saturation Structure

Most systems show a three-phase structure:

1. Low-dimensional local regime
2. Growth phase with increasing dimensionality
3. Saturation phase with bounded dimensionality

3.3 Cross-Domain Alignment

After normalization, dimensionality curves align strongly across systems (Fig 2), with mean similarity ≈ 0.90.

3.4 TRIBE-Specific Compression

TRIBE representations show:

- Low effective dimensionality ($D_{\text{eff}} \approx 3$–5)
- Early saturation (Fig 1)
- High compression ratio (≈ 0.3)

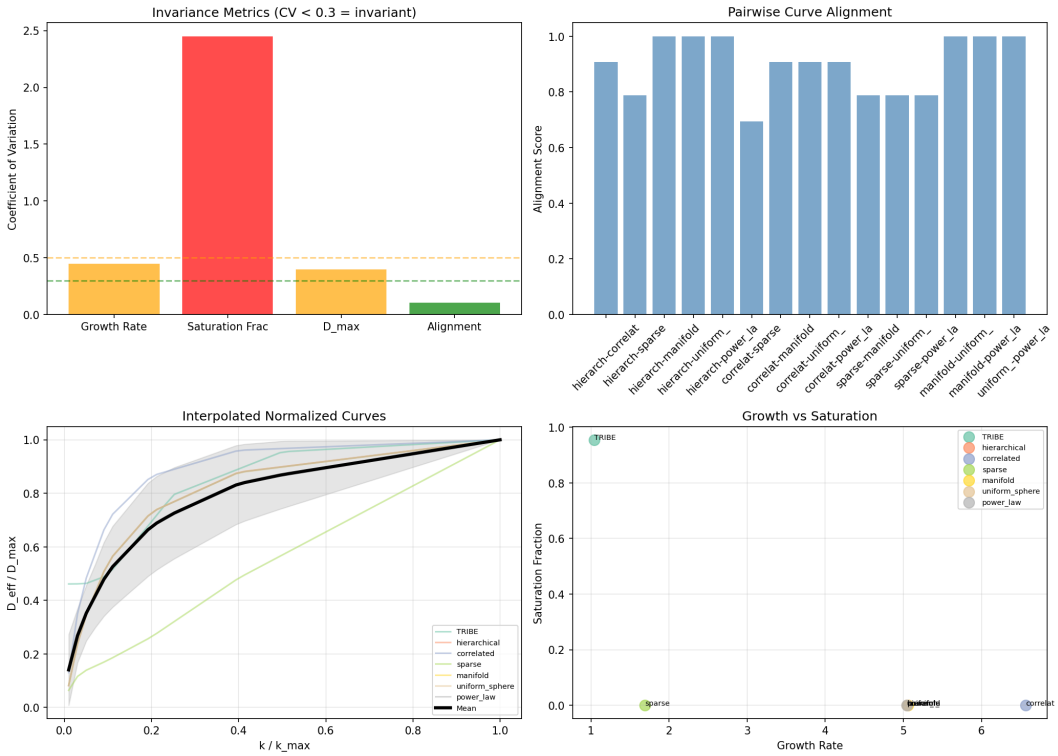

Figure 2: Normalized dimensionality curves across systems. Despite differences in scale and magnitude, curves exhibit strong alignment, indicating a shared structural motif.

3.5 Negative Results from Alternative Metrics

Dynamical metrics including torsion (χ), curvature (κ), and divergence failed to show robust alignment across dynamical regimes (Fig 3).

3.6 Cross-Domain Saturation Comparison

Growth–saturation structure is observed in most systems, though saturation levels differ substantially (Fig 4).

4 Discussion

4.1 A Shared Structural Motif

These results identify a consistent structural pattern:

> Effective dimensionality increases with scale and saturates at a bounded level.

This pattern is consistent with hierarchical organization and constrained representational systems.

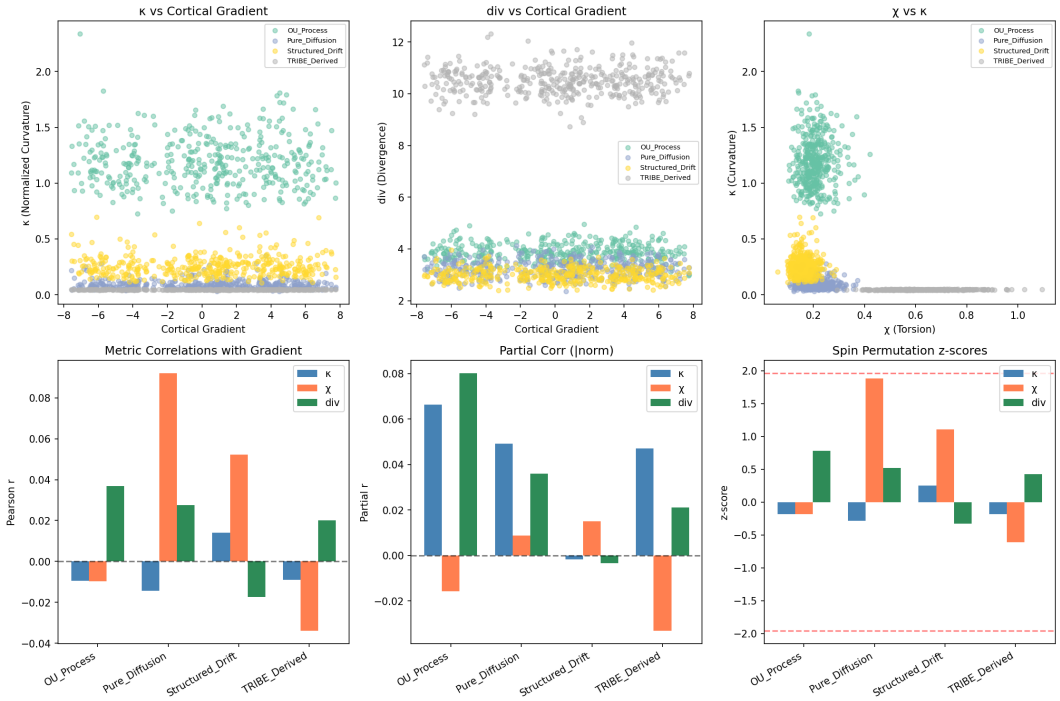

Figure 3: Comparison of dynamical and statistical metrics across systems. None show consistent or robust alignment with structural organization, in contrast to effective dimensionality.

4.2 Quantitative Differences

Systems differ in:

- growth rate
- saturation level
- compression strength

TRIBE exhibits unusually strong compression and early saturation.

4.3 Negative Results and Robustness

Alternative metrics fail to replicate these patterns, strengthening the specificity of the dimensionality-based analysis.

4.4 Limitations

- Limited real-world datasets
- Dependence on model-derived representations
- No causal mechanism established

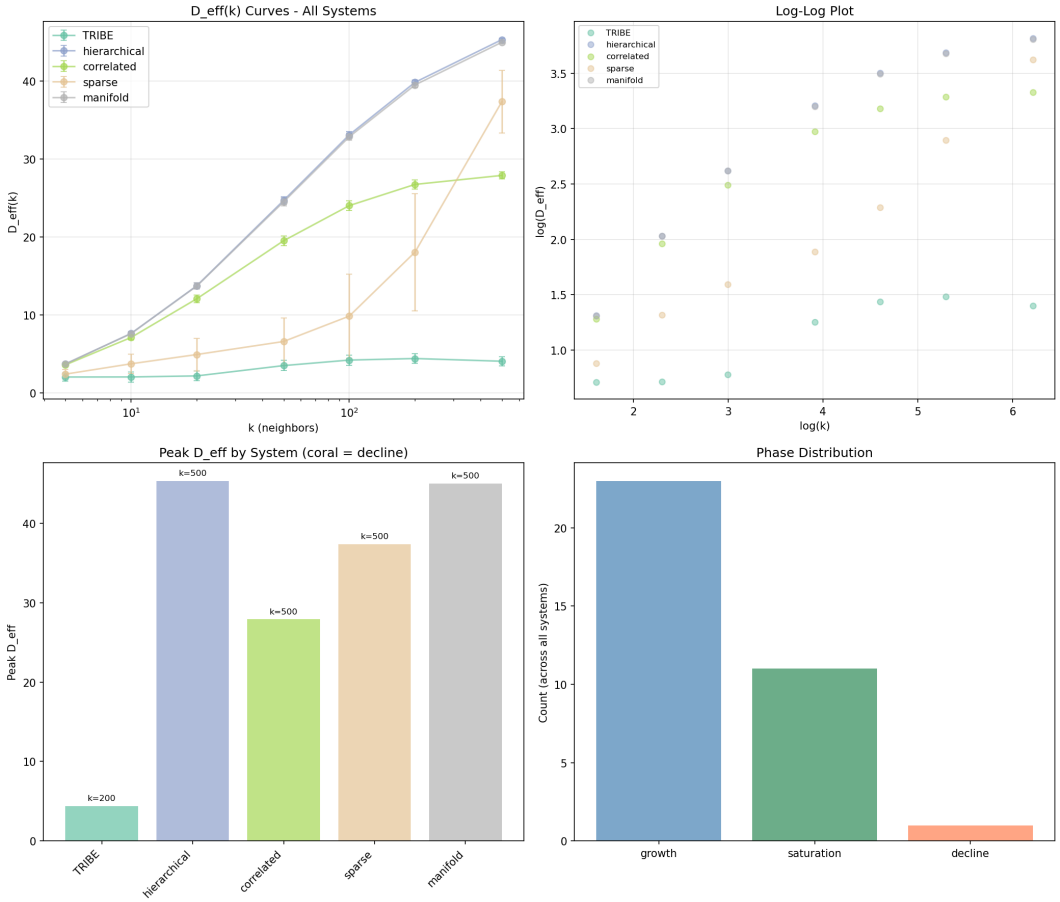

Figure 4: Cross-domain comparison of dimensionality scaling. Most systems exhibit growth–saturation structure, though quantitative parameters differ.

5 Conclusion

We identify scale-dependent dimensional growth with bounded saturation as a common structural property of representational systems. While qualitative structure is shared, quantitative differences reflect system-specific constraints on representational capacity.

Data Availability

`https://github.com/urmt/SGP-Tribe3`

References

Daniel S Margulies et al. Situating the default-mode network along a principal gradient of macroscale cortical organization. *Proceedings of the National Academy of Sciences*, 113(44): 12574–12579, 2016.